

25. (Singular Value Decomposition) Suppose $A = UDV^T$, where U and V are $n \times n$ matrices with the property that $U^T U = I$ and $V^T V = I$, and where D is a diagonal matrix with positive numbers $\delta_1, \dots, \delta_n$ on the diagonal. Show that A is invertible and find a formula for A^{-1} .

Solution: We shall first show that D is invertible.

Let E_1 be the $n \times n$ matrix whose Row 2 to Row n are equivalent to I_n .

Row 1 of $E_1 = \frac{1}{\delta_1} \times \text{Row 1 of } I_n$.

Let E_2 be the $n \times n$ matrix whose Rows apart from 2 are equivalent to I_n .

Row 2 of $E_2 = \frac{1}{\delta_2} \times \text{Row 2 of } I_n$.

⋮ This continues for up to E_n .

Let E_n be the $n \times n$ matrix whose Rows apart from n are equivalent to I_n .

Row n of $E_n = \frac{1}{\delta_n} \times \text{Row } n \text{ of } I_n$.

The Diagonal matrix D can be converted to I_n by multiplying each row i of D by $\frac{1}{\delta_i}$, where $1 \leq i \leq n$.

By Prob 24(c) of Exercise 2.2, $(E_n E_{n-1} \dots E_2 E_1) D = I$.

$\therefore D$ is row equivalent to I . By Theorem 7, D is invertible.

By Defn. of Transpose, U^T and V^T are both $n \times n$ matrices.

By Prob 25 from Exercise 2.3, U and V are both invertible and $U^{-1} = U^T$ and $V^{-1} = V^T$.

$$A = UDV^T \Rightarrow U^T A = U^T (UDV^T) \Rightarrow U^T A = U^T (U(DV^T)) \quad (\text{Theorem 2(a)})$$

$$\Rightarrow U^T A = (U^T U)(DV^T) \quad (\text{Theorem 2(a)})$$

$$\Rightarrow U^T A = I(DV^T) \quad (\text{Given})$$

$$\Rightarrow U^T A = DV^T \quad (\text{Theorem 2(e)}) \Rightarrow D^{-1}(U^T A) = D^{-1}(DV^T) \quad (\text{We have proved that } D^{-1} \text{ exists})$$

$$\Rightarrow (D^{-1}U^T)A = (D^{-1}D)V^T \quad (\text{Theorem 2(a)})$$

$$\Rightarrow (D^{-1}U^T)A = IV^T \quad (\text{By Defn. of inverse})$$

$$\Rightarrow (D^{-1}U^T)A = V^T \quad (\text{Theorem 2(e)})$$

$$\Rightarrow V(D^{-1}U^T)A = VV^T \Rightarrow V(D^{-1}U^T)A = VV^{-1} \quad (\text{Proved earlier that } V^T = V^{-1})$$

$$\Rightarrow (V(D^{-1}U^T))A = VV^{-1} \quad (\text{Theorem 2(a)}) \Rightarrow ((VD^{-1})U^T)A = VV^{-1} \quad (\text{Theorem 2(a)})$$